



HIGHLY CONCENTRATED

A laser is a device that emits light through the process of optical amplification, based on stimulated emission (see opposite). Unlike light from other sources, laser light is coherent, meaning that the waves are perfectly in step with each other and have the same frequency. The invention of lasers has allowed unprecedented control over light, with applications such as remote-sensing lidar, laser cutting, and spectroscopy. Lasers can also be used to trap and cool small particles, such as atoms and ions.

RUBY LASER

**HIGH-INTENSITY
FLASH LAMP**

The ruby medium is pumped using a flashtube. Energy emitted from the flashtube is absorbed by electrons in the medium, which then jump to excited states.

ALL WAVES IN STEP

The laser emits rapid pulses of visible red light. Due to the photons being produced via stimulated emission, this light is perfectly coherent.